

**Directions:** Students MUST sign-up (by Friday, February 27<sup>th</sup> @ 8pm eastern) for an oral exam time between Monday, March 2<sup>nd</sup> and Wednesday, March 4<sup>th</sup>. You will sign-up for a 30-minute timeframe (though we will expect a 20-minute oral exam).

Students may collaborate with (only) their classmates to prepare for the oral exam, but only until Monday, March 2<sup>nd</sup> at 10:00 AM, when the oral exams may begin. Each student is allowed to bring in two pieces of 8.5" by 11" paper with notes (handwritten) to refer to, as well as a calculator. (Dr. Herman will provide the formula sheet from the in-class exam as well). (You may also ask Dr. Herman questions). Canvas is the only website a student should access for help with the exam.

Students will be asked questions based on two of the following questions. Students will get to choose one of the questions she will base questions on while Dr. Herman will choose the other. Students may be doing a mix of writing (while explaining) and just explaining. Note – this will likely not just be “working the problem” for me.

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**Question 1:** A company has a certain allergy treatment on the market (cost is \$10/month) already but has recently developed a new treatment (cost will be \$50/month). The variable of interest is how long it takes for the allergy sufferer to start to feel relief after taking the medication. We would like to see if the new treatment is significantly better than the current treatment. To help determine this, a random sample of 82 allergy sufferers is taken and 41 are randomly assigned to each treatment. It is then measured (with consultation by the allergy experts) how long it takes for the relief to start, measured in minutes. Summary statistics are given in the table below:

|          | Treatment |         |
|----------|-----------|---------|
|          | New       | Current |
| Mean     | 42.45     | 46.21   |
| Std. Dev | 10.78     | 7.21    |
| n        | 41        | 41      |

- Determine if it is feasible to assume that the population variances are equal for the two medications. Use a significance level of 0.05. Be sure to show all steps!!
- Using the appropriate t-test, determine if the new treatment is significantly better than the current. Use a significance level of 0.05.
- Given the results of your tests, make a recommendation as to which medication the company should market, assuming they could only pick one. Defend your choice using results from (a) and (b) as well as practical implications.

**Question 2:** Recall: The exponential distribution has one parameter  $\beta$ , which is its expected value. Suppose we are interested in the population median of the exponential distribution that we will call  $m$ .

- Using the exponential pdf, find its median,  $m$  (which will be a function of  $\beta$ ).
- Note that  $E(\bar{Y}) = \beta$ . Thus, in the equation you found in part (a) for the median, substituting  $\bar{Y}$  for  $\beta$  would be a reasonable estimator of  $m$ . Call this  $\hat{m}$ . Find  $E(\hat{m})$  and  $Var(\hat{m})$ .
- Using the mean and variance that you found in part (b), construct a z-score which could be used to test the null hypothesis  $H_0: m = m_0$ . (FYI - this only works due to the Delta Method)
- Use the z-score you created in part (c) to test  $H_0: m = 1.8$  vs.  $H_a: m > 1.8$  using the following data (5% significance level). Be sure you show all steps!

|     |     |     |     |     |      |     |     |     |      |     |     |     |
|-----|-----|-----|-----|-----|------|-----|-----|-----|------|-----|-----|-----|
| 6.4 | 3.5 | 3.2 | 3.7 | 4.9 | 0.7  | 0.9 | 0.2 | 1.6 | 18.6 | 3.7 | 2.4 | 2.1 |
| 0.6 | 1.3 | 3.7 | 0.1 | 4.0 | 1.6  | 0.3 | 0.3 | 0.8 | 7.9  | 6.6 | 0.4 | 0.5 |
| 1.3 | 6.5 | 3.7 | 4.9 | 0.8 | 10.8 | 2.0 | 2.9 | 1.8 | 4.6  | 2.0 | 2.5 | 1.4 |
| 3.9 | 0.2 | 1.4 | 3.9 | 2.6 | 7.3  | 0.6 | 5.1 | 0.8 | 2.0  | 2.6 | 3.6 | 9.1 |

**Question 3:** If a 95% confidence interval for  $\sigma^2$  is reported to be (28.44, 82.603), what was the value of the sample standard deviation? To find the value of  $s$ , you will also have to find the value of the sample size.

**Question 4:** In October of 2024, it was estimated that 13% of U.S. households experience food insecurity. We would like to see if this value has increased since then.

- If we take a random sample of 400 U.S. households and if we reject  $H_0$  if 60 or more of the households experience food insecurity, what would be the significance level of the test?
- What is the power of the test if in reality, 15.8% of U.S. households experience food insecurity?
- Looking at the power calculated in part b, suppose I would like this power to be 80%. How big of a sample would we need to take to achieve this power?

Note, you should also be able to draw the pictures and should be able to tell me how values would change based on changing other values.

**Question 5:** Suppose you have 10 observations,  $Y_1, Y_2, \dots, Y_{10}$  that come from a normal population with mean of 2 and a standard deviation of 3. Using ONLY these  $Y$ 's and their sample mean (no  $S^2$  allowed) find the following: (Note – be sure to explain WHY it would have that distribution)

- A standard normal random variable.
- A chi-square random variable with 4 degrees of freedom.
- A T random variable with 7 degrees of freedom
- An F random variable with 1 numerator and 7 denominator degrees of freedom.